Range

1

a 13 **b** 31 **c** 2.2981

2

a 5 **b** 72

3 4

4 23

5 11

6

a One person in the group **b** 15 people

c 16 names **d** 10 names

e 6

Interquartile range

7

a

i 7 **ii** 13 **iii** 15 **iv** 7

v 16 **vi** 9

b

i 7 **ii** 1 **iii** 18 **iv** -3

v 9 **vi** 12

c

i 7 **ii** 38 **iii** 38 **iv** 33

v 48 **vi** 15

d

i 7 **ii** 2.9 **iii** 3.6 **iv** 1.6

v 3.5 **vi** 1.9

e

i 15 **ii** 35 **iii** 27 **iv** 27

v 40 **vi** 13

f

i 19 **ii** 28 **iii** 43 **iv** 13

v 38 **vi** 25

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8

a

i 15 ii 9 iii 16 iv 7

b

i 17.5 ii 45 iii 27.5 iv 16

c

i 25 ii 17.5 iii 45 iv 27.5

d

i 25 ii 19.5 iii 26.5 iv 7

9

a

i 9 ii 10 iii 1

b

i 1 ii 3 iii 2

c

i 72 ii 85 iii 13

10

a $15 **b** $16.50 **c** International

11

a 47 **b** 7 **c** 4

12

a 51 **b** 7 **c** 1

13

a

Class	Class center	Frequency
$1-5$	3	5
$6-10$	8	7
$11-15$	13	6
$16-20$	18	10

b

i 15 ii 10

14

a 75 **b** 8 **c** 3

15 8.8

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16 50

17

a

Ball speed (km/h)	Class center	Frequency
$110 \leq x < 120$	115	4
$120 \leq x < 130$	125	12
$130 \leq x < 140$	135	8
$140 \leq x < 150$	145	5
$150 \leq x < 160$	155	2

b

i 40 **ii** 125 **iii** 135 **iv** 10

18

a

Height (cm)	Class center	Frequency
$95 \leq x < 100$	97.5	1
$100 \leq x < 105$	102.5	3
$105 \leq x < 110$	107.5	2
$110 \leq x < 115$	112.5	12
$115 \leq x < 120$	117.5	18
$120 \leq x < 125$	122.5	15
$125 \leq x < 130$	127.5	4
$130 \leq x < 135$	132.5	4

b

i 35 **ii** 10

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19

a

i 86 **ii** 109 **iii** 23 **iv** 96

v 107 **vi** 11

b

Weight (g)	Class center	Number of apples
$85 \leq x < 90$	87.5	2
$90 \leq x < 95$	92.5	3
$95 \leq x < 100$	97.5	5
$100 \leq x < 105$	102.5	7
$105 \leq x < 110$	107.5	13

c

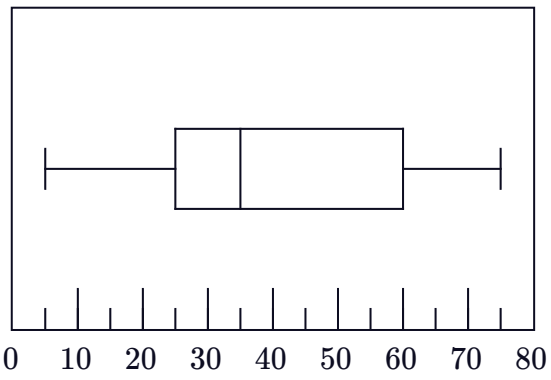
i 20 **ii** 10

d 1

Box plots

20 7

21

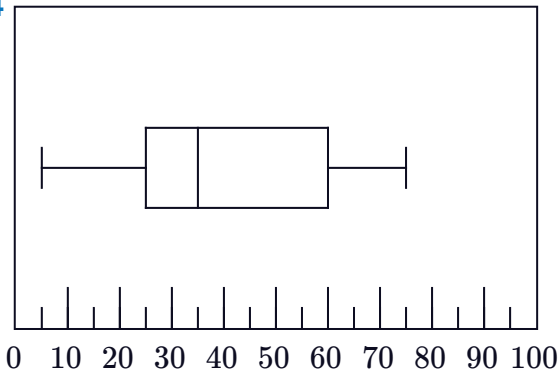


22 33

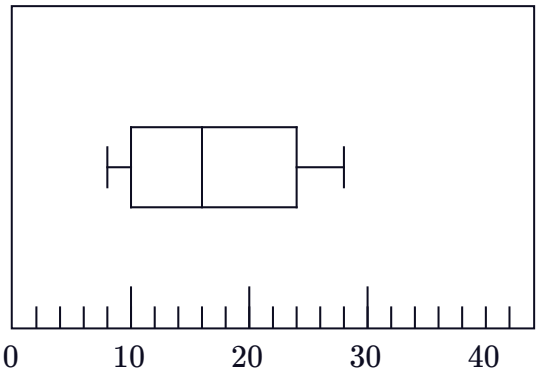
23 42

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24



25

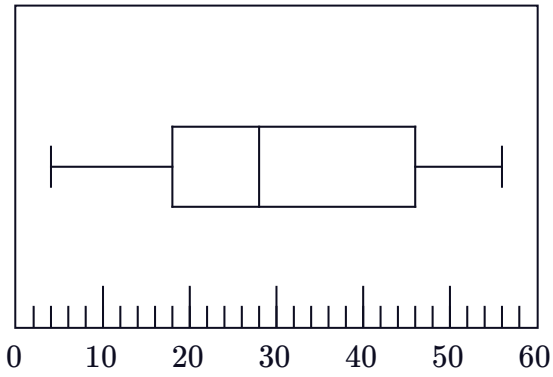


26

a

Minimum	4
Lower Quartile	18
Median	28
Upper Quartile	46
Maximum	56

b



27

a 18.97 kg

b

Q1	17 kg
Median	19 kg
Q3	21 kg

c 2nd

28

a 3 **b** 18 **c** 15 **d** 10

e 7

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29

a

i 50% ii 25% iii 50% iv 75%

v 75%

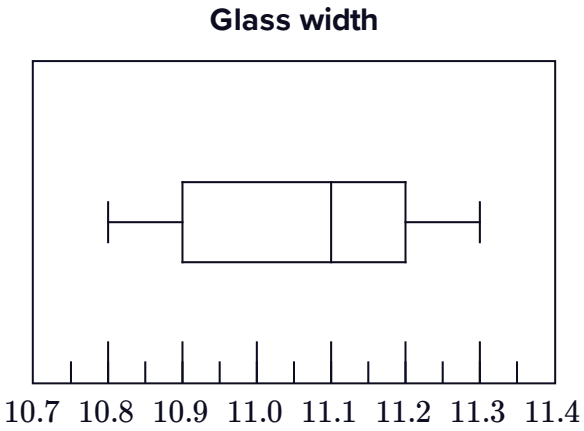
b 2nd

30

a 11.10 mm

b 0.3 mm

c



d 78.95%

e 2nd

f The mode and mean

31 No. For less than 25% of the time, he blacks out for longer than 7 seconds.

32

a 31 years **b** 17 years **c** 25% **d** 7

e 1st **f** 5

Standard deviation

33

a

i 5.30 ii 5.58

b

i 9.57 ii 10.15

c

i 2.01 ii 2.02

d

i 18.74 ii 19.51

e

i 1.58 ii 1.59

f

i 1.58 ii 1.58

g

i 1.61 ii 1.65

h

i 1.36 ii 1.38

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34

a 0.56

b

i Increase **ii** Increase

c 0

35

a 0.57

b Decrease the mean and increase the standard deviation.

36 Decrease both the mean and standard deviation.

37

a 12

b 9

c 4.7

d Standard deviation. For an accurate representation of spread, every score should be taken into account.

38 The mean for Paper 1 would be greater and the standard deviation would be smaller than that of Paper 2.

39

a $\$5404$ million **b** greater **c** greater

40

a 16

b 9

c

Score (x)	$x-\text{mean}$	$(x-\text{mean})^2$
19	3	9
18	2	4
14	-2	4
19	3	9
10	-6	36

d 3.94

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41

a 11

b 56

c

Score (x)	$x - \text{mean}$	$(x - \text{mean})^2$
-15	-26	676
-7	-18	324
10	-1	1
26	15	225
41	30	900

d 23.05

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